**EXPERIMENT 7: - MODEL PLANNING AND BUILDING – CLUSTERING**

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**Learning Objective**

The objective of this experiment is to enable students to understand and implement the K-Means clustering algorithm for customer segmentation.

**Problem Statement**

A retail company wants to group its customers based on their Annual Income and Spending Score to design personalized marketing campaigns. By identifying customers with similar purchasing behavior, the company can improve customer satisfaction, optimize promotional offers, and increase overall sales.

Your task is to analyze the customer dataset, apply the K-Means Clustering algorithm to identify customer segments, visualize the clusters, and recommend suitable marketing strategies for each segment.

**Tasks to Perform**

1. Use the Mall Customer Segmentation dataset from Kaggle and load it into a Pandas DataFrame.
2. Explore the dataset by displaying the first and last five records, checking the shape and data types, identifying missing values and duplicates, and displaying summary statistics.
3. Select Annual Income and Spending Score as clustering features.
4. Standardize the selected features using StandardScaler.
5. Visualize the selected features using scatter plots and pair plots.
6. Determine the optimal number of clusters using the Elbow Method and plot the Elbow Curve.
7. Train the K-Means clustering model using the selected number of clusters, assign cluster labels, and save them in a new column.
8. Visualize the clusters using a scatter plot, color the clusters, mark the centroids, and display their coordinates.
9. Calculate the number of customers in each cluster.
10. Calculate the average Annual Income and Spending Score for each cluster.

**AIM**

To perform customer segmentation using the K-Means clustering algorithm based on Annual Income and Spending Score.

**ALGORITHM**

1. Import the required libraries and load the Mall Customer Segmentation dataset.
2. Display the first and last five records, dataset shape, data types, missing values, duplicates, and summary statistics.
3. Select Annual Income and Spending Score as features.
4. Standardize the selected features using StandardScaler.
5. Visualize the features using scatter plots and pair plots.
6. Use the Elbow Method to determine the optimal number of clusters.
7. Apply K-Means clustering and assign cluster labels to customers.
8. Visualize the clusters using a scatter plot with centroids.
9. Calculate the number of customers in each cluster.
10. Calculate the average Annual Income and Spending Score for each cluster.

**PROGRAM AND OUTPUT**

## Question 1: Load the Mall Customer Segmentation dataset from an online CSV into a Pandas DataFrame.

```
# LUQMAN T (241801141)

import pandas as pd

url = "https://raw.githubusercontent.com/DataThinkers/Datasets/main/DS/Mall_Customers.csv"
df = pd.read_csv(url)
print(df.head())
```

```
   CustomerID   Genre  Age  Annual Income (k$)  Spending Score (1-100)
0           1    Male   19                  15                      39
1           2    Male   21                  15                      81
2           3  Female   20                  16                       6
3           4  Female   23                  16                      77
4           5  Female   31                  17                      40
```

## Question 2: Explore the dataset by displaying first and last five records, shape, data types, missing values, duplicates, and summary statistics.

```
# LUQMAN T (241801141)

print(df.head())
print(df.tail())
print("Shape:", df.shape)
print(df.dtypes)
print("Missing values:\n", df.isnull().sum())
print("Duplicates:", df.duplicated().sum())
print(df.describe(include="all"))
```

```
   CustomerID   Genre  Age  Annual Income (k$)  Spending Score (1-100)
0           1    Male   19                  15                      39
1           2    Male   21                  15                      81
2           3  Female   20                  16                       6
3           4  Female   23                  16                      77
4           5  Female   31                  17                      40
     CustomerID   Genre  Age  Annual Income (k$)  Spending Score (1-100)
195         196  Female   35                 120                      79
196         197  Female   45                 126                      28
197         198    Male   32                 126                      74
198         199    Male   32                 137                      18
199         200    Male   30                 137                      83
Shape: (200, 5)
CustomerID                int64
Genre                    object
Age                       int64
Annual Income (k$)        int64
Spending Score (1-100)    int64
dtype: object
Missing values:
 CustomerID              0
Genre                    0
Age                      0
Annual Income (k$)       0
Spending Score (1-100)   0
dtype: int64
Duplicates: 0
        CustomerID   Genre         Age  Annual Income (k$)  \
count   200.000000     200  200.000000          200.000000
unique         NaN       2         NaN                 NaN
top            NaN  Female         NaN                 NaN
freq           NaN     112         NaN                 NaN
mean    100.500000     NaN   38.850000           60.560000
std      57.879185     NaN   13.969007           26.264721
min       1.000000     NaN   18.000000           15.000000
25%      50.750000     NaN   28.750000           41.500000
50%     100.500000     NaN   36.000000           61.500000
75%     150.250000     NaN   49.000000           78.000000
max     200.000000     NaN   70.000000          137.000000

        Spending Score (1-100)
count               200.000000
unique                     NaN
top                        NaN
freq                       NaN
mean                 50.200000
```

```
std                 25.823522
min                  1.000000
25%                 34.750000
50%                 50.000000
75%                 73.000000
max                 99.000000
```

## Question 3: Select Annual Income (k$) and Spending Score (1-100) as features for clustering.

```
# LUQMAN T (241801141)

X = df[["Annual Income (k$)", "Spending Score (1-100)"]]
print(X.head())
```

```
   Annual Income (k$)  Spending Score (1-100)
0                  15                      39
1                  15                      81
2                  16                       6
3                  16                      77
4                  17                      40
```

## Question 4: Standardize the selected features using StandardScaler.

```
# LUQMAN T (241801141)

from sklearn.preprocessing import StandardScaler

scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)
print(X_scaled[:5])
```

```
[[-1.73899919 -0.43480148]
 [-1.73899919  1.19570407]
 [-1.70082976 -1.71591298]
 [-1.70082976  1.04041783]
 [-1.66266033 -0.39597992]]
```

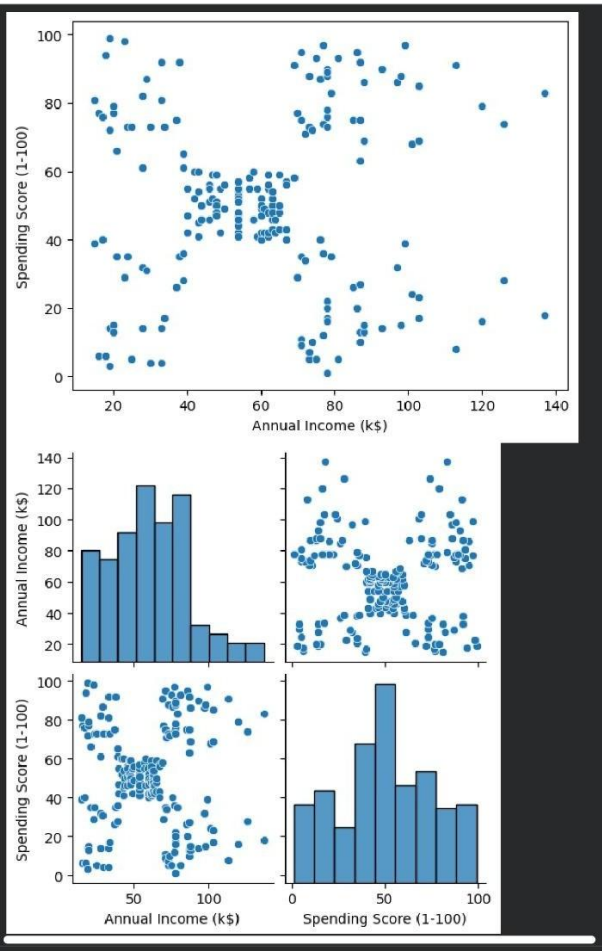
## Question 5: Visualize the selected features using a scatter plot and pair plot.

```
# LUQMAN T (241801141)

import seaborn as sns
import matplotlib.pyplot as plt

sns.scatterplot(data=df, x="Annual Income (k$)", y="Spending Score (1-100)")
plt.show()

sns.pairplot(X)
plt.show()
```



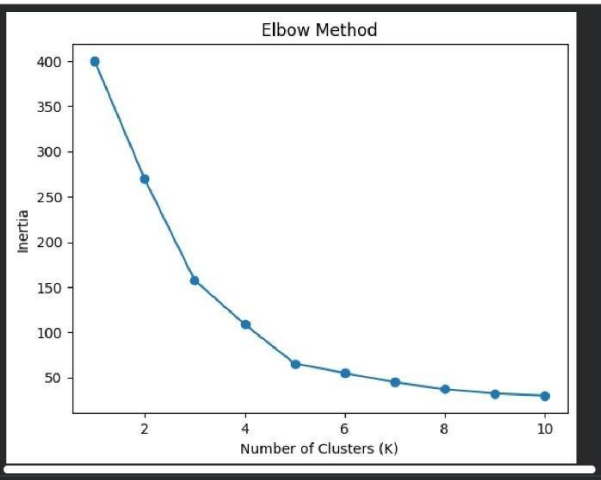

Question 6: Determine the optimal number of clusters using the Elbow Method and plot the Elbow Curve.

```
# LUQMAN T (241801141)

from sklearn.cluster import KMeans

inertia = []
for k in range(1, 11):
    km = KMeans(n_clusters=k, random_state=42, n_init=10)
    km.fit(X_scaled)
    inertia.append(km.inertia_)

plt.plot(range(1, 11), inertia, marker="o")
plt.xlabel("Number of Clusters (K)")
plt.ylabel("Inertia")
plt.title("Elbow Method")
plt.show()
```




## Question 7: Train K-Means using the optimal number of clusters, assign labels, and save the dataset to a file.

```
# LUQMAN T (241801141)

k = 5
kmeans = KMeans(n_clusters=k, random_state=42, n_init=10)
df["Cluster"] = kmeans.fit_predict(X_scaled)

df.to_csv("mall_customers_clustered.csv", index=False)
print(df.head())
```

```
   CustomerID   Genre  Age  Annual Income (k$)  Spending Score (1-100)  \
0           1    Male   19                  15                      39
1           2    Male   21                  15                      81
2           3  Female   20                  16                       6
3           4  Female   23                  16                      77
4           5  Female   31                  17                      40

   Cluster
0        4
1        2
2        4
3        2
4        4
```

## Question 8: Visualize customer clusters, highlight centroids, and display centroid coordinates.

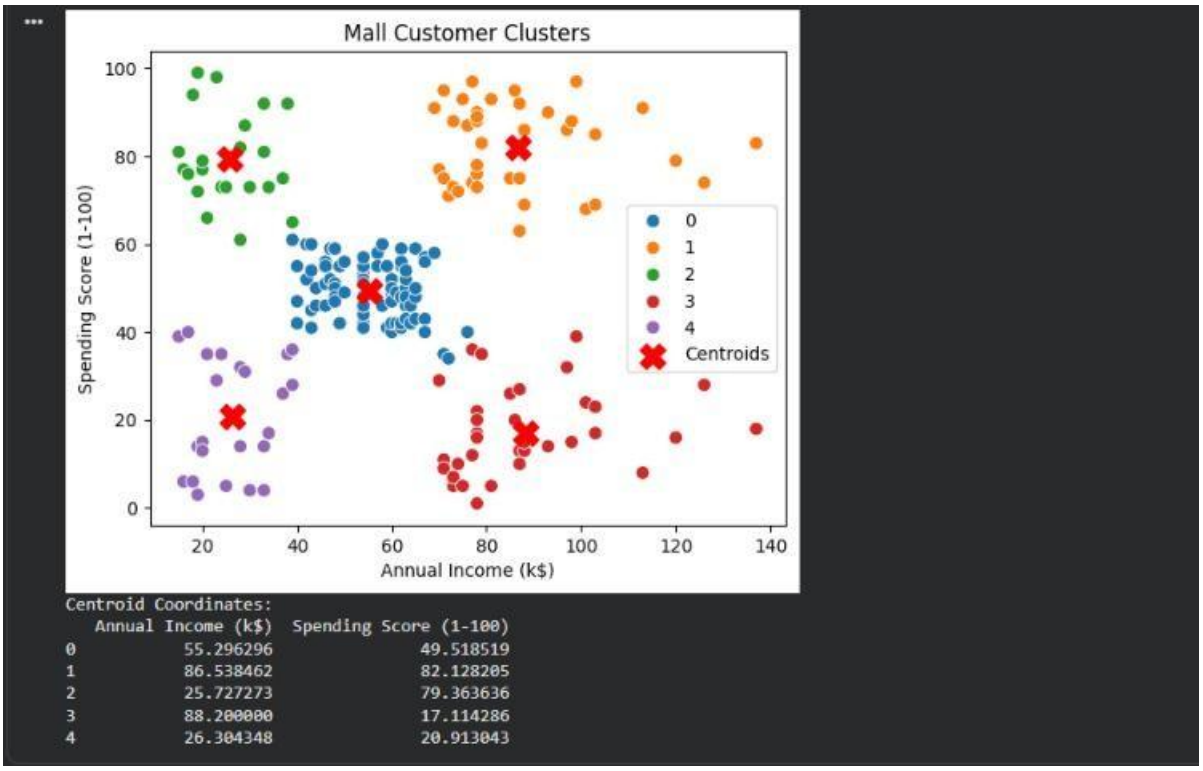
```
# LUQMAN T (241801141)

centers = scaler.inverse_transform(kmeans.cluster_centers_)

sns.scatterplot(
    data=df, x="Annual Income (k$)", y="Spending Score (1-100)",
    hue="Cluster", palette="tab10", s=60
)

plt.scatter(centers[:, 0], centers[:, 1], c="red", marker="X",
            s=200, label="Centroids")
plt.title("Mall Customer Clusters")
plt.legend()
plt.show()

print("Centroid Coordinates:")
print(pd.DataFrame(centers, columns=X.columns))
```




```
Centroid Coordinates:
   Annual Income (k$)  Spending Score (1-100)
0           55.296296               49.518519
1           86.538462               82.128205
2           25.727273               79.363636
3           88.200000               17.114286
4           26.304348               20.913043
```

Question 9: Calculate the number of customers in each cluster.

```
# LUQMAN T (241801141)

print(df["Cluster"].value_counts().sort_index())
```

```
Cluster
0    81
1    39
2    22
3    35
4    23
Name: count, dtype: int64
```

Question 10: Calculate the average Annual Income and Spending Score for each cluster.

```
# LUQMAN T (241801141)

print(df.groupby("Cluster")[[
    "Annual Income (k$)", "Spending Score (1-100)"
]].mean())
```

```
         Annual Income (k$)  Spending Score (1-100)
Cluster
0                 55.296296               49.518519
1                 86.538462               82.128205
2                 25.727273               79.363636
3                 88.200000               17.114286
4                 26.304348               20.913043
```

**RESULT**

The K-Means clustering algorithm was applied to segment customers based on annual income and spending score. The clusters were visualized and analyzed to identify customer groups for targeted marketing strategies.